Rahul Biswas

cs22resch01004@iith.ac.in | 007biswasrahul@gmail.com | +91 7384357065 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2026	8.50
Hijli High School	Class XII (West Bengal Council of Higher Secondary Education (WBCHSE))	2012	75.20%
Hijli High School	Class X (West Bengal Board of Secondary Education (WBBSE))	2010	84.37%

PUBLICATIONS

- **4 research papers in international conference proceedings: 3 accepted** for the ***IEEE International Joint Conference on Neural Networks (IJCNN)*** and **1 published** in the ***International Conference on Pattern Recognition (ICPR)***.
- *"Task-Free Online Replay with Contrastive Learning and Dynamic Herding."* -- ***ICPR 2026***. ***Published***. **Rahul Biswas**, C. Krishna Mohan, Subhrajit Nag, Sandipan Dandapat.
- *"Efficient Proxy-Based Memory Management for Online Federated Continual Learning."* ***IJCNN 2026***. ***Accepted***. **Rahul Biswas**, Aditya Varun V, C. Krishna Mohan.
- *"GAKD: Geometry-Aware Knowledge Distillation for Peer-to-Peer Federated Learning."* ***IJCNN 2026***. ***Accepted***. **Rahul Biswas**, Supreet Shukla, C. Krishna Mohan.
- *"FedGCR: Federated Open Set Recognition Using Gaussian Noises in Client's Representation."* ***IJCNN 2026***. ***Accepted***. Ankita Das, Mrinmay Sen, **Rahul Biswas**, C. Krishna Mohan.
- **Papers Under Review**
- *"COMET: Confidence-Guided Memory Management for Decentralized Online Federated Continual Learning." Special Issue on Advances in Federated Learning for Critical Applications.* **Under Review**. **Rahul Biswas**, Mrinmay Sen, C. Krishna Mohan.
- *"FAGH: Accelerating Federated Learning with Approximated Global Hessian." IEEE Transactions on Neural Networks and Learning Systems (TNNLS).* **Under Review**. Mrinmay Sen, A. Qin, **Rahul Biswas**, C. Krishna Mohan.
- *"SPADE: Sparse Prototype-Anchored Concept Bottleneck Model for Continual Learning." AAAI Conference on Artificial Intelligence (AAAI), 2027.* **Under Review**. **Rahul Biswas**, Prashant Kumar Dubey, Mrinmay Sen, C. Krishna Mohan.

INSTITUTE BUILD PROJECT

Institute BUILD Project -- Autonomous Drone-Based Medicine Delivery (January-April, 2022)

- Developed a *computer-vision-based obstacle avoidance module* for an autonomous drone designed to deliver medicines from the Technology Hospital to hostel areas.
- Explored *depth estimation techniques* to derive spatial information from camera imagery and enable the drone to perceive its surrounding environment.
- Integrated depth-based scene understanding with *obstacle detection* to identify potential collisions and support safer flight decisions.
- Focused on developing a *practical and deployable solution*, considering real-world requirements such as computational efficiency and real-time processing for autonomous drone operation.

POSITIONS OF RESPONSIBILITY

- **Conference Reviewer:** *IEEE International Joint Conference on Neural Networks (IJCNN), 2025, 2026*.
- **Sub-Reviewer:** *Asian Conference on Machine Learning (ACML)*, 2025; *British Machine Vision Conference (BMVC), 2025*.

M.TECH THESIS

Brain Tumor Classification using Deep CNN Architectures via Transfer Learning

- Developed a *CNN-based brain tumor classification system* for MRI images to classify *glioma, meningioma,* and *pituitary tumors*, with emphasis on automated feature extraction and accurate multi-class classification.
- Applied *transfer learning using pretrained CNN architectures* to leverage learned visual representations and improve classification performance on a limited medical imaging dataset.
- Investigated and compared multiple deep CNN architectures, including *AlexNet, VGG, MobileNet,* and *EfficientNet*, to evaluate their suitability for brain tumor classification.
- Developed an end-to-end pipeline involving *MRI image preprocessing, normalization, resizing, data augmentation, transfer learning, feature extraction, and tumor classification*.
- Explored *Knowledge Distillation* to transfer knowledge from a high-capacity teacher model to a lightweight student model, targeting a more computationally efficient model while retaining classification performance.

SKILLS

- **Machine Learning & Deep Learning:** Federated Learning, Continual Learning, Federated Continual Learning, Representation Learning, Contrastive Learning, Knowledge Distillation, Open Set Recognition, Computer Vision, Vision-Language Models, Concept Bottleneck Models.
- **Programming Languages:** Python, C++.
- **Frameworks & Libraries:** PyTorch, PyTorch Lightning, Hugging Face Transformers, OpenCLIP, timm, Torchvision, OpenCV, NumPy, SciPy, Pandas, Matplotlib, Scikit-learn.
- **Infrastructure & Tools:** CUDA, Distributed GPU Clusters, Slurm (HPC), Git, Linux, Windows.
- **Academic Skills:** Academic Writing, Peer Reviewing, Teaching, Scientific Communication.

RELEVANT COURSES

- Deep Learning, Machine Learning, Artificial Intelligence, Visual Computing, Algorithms and Data Structures.

EXTRACURRICULAR

- **Organizing Committee Member:** *Indo-US Workshop on Harnessing AI for Healthcare Transformation (HAHT-IUS)*, 2024.
- **NSS Volunteer:** Participated in the *Swachh Bharat (Clean India) Drive, 2026*, contributing to community cleanliness and awareness initiatives.